CSIT321 · UNIVERSITY OF WOLLONGONG IN DUBAI · TECHCODEX

OptiRide

PREDICT | OPTIMIZE | PROTECT

AI-Powered Driver Allocation & Safety Monitoring for Last-Mile Food Delivery

TEAM TECHCODEX

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PROBLEM & CONTEXT

The UAE food delivery sector is booming — but behind every order is a driver facing extreme heat, fatigue, and dangerous roads. Existing platforms rely on reactive dispatch: drivers are over-extended, idle, and unmonitored.

Reactive Dispatch

Drivers assigned only after orders arrive — no predictive deployment, leading to idle time and missed surges

Zero Safety Net

No real-time fatigue or accident detection — UAE has recorded hundreds of delivery rider road incidents

HOW OPTIRIDE RESPONDS

- Predict demand surges per zone before they happen using ML
- Detect driver fatigue & accidents instantly via IoT sensors
- Deliver personalised AI insights to drivers & fleet managers
- Deploy idle drivers proactively to high-demand areas
- Reduce idle kilometres, fuel consumption & CO_2 emissions

CORE FEATURES

Predictive Demand Forecasting

Random Forest Regressor, Gradient Boosting & Ridge Regression predict zone-level surges using live traffic & weather data

Smart Driver Allocation

DBSCAN clustering dynamically groups zones and deploys the nearest idle driver

Fatigue & Safety Detection

MediaPipe detects fatigue via face scanning; gyro & accelerometer and gyroscope flag collisions with instant alerts

Generative AI Insights

Gemini AI produces personalised operational tips, route summaries & safety recommendations

Route Optimisation

Google Maps API delivers traffic-aware routing in real time, minimising distance & emissions

SYSTEM ARCHITECTURE

CLIENT SIDE

- **Admin Web Dashboard** — React · Vite
- **Driver Mobile App** — React Native · Expo

AUTH & GATEWAY

- **Firebase Auth** — Secure login · RBAC
- **API Gateway (FastAPI)** — Routes all client requests to services

Apache Kafka — Real-Time Streaming

AI & PROCESSING MODULES

- **Demand Forecast** — RF Regressor · Gradient Boosting · Ridge
- **Allocation & Routing** — DBSCAN
- **Safety Monitor** — MediaPipe · IoT Sensors
- **AI Insights** — Gemini API · Cache

DATA LAYER

- **PostgreSQL + PostGIS** — Routes · Logs · Locations
- **Redis** — Cache · DB lock
- **Google Cloud Storage** — Data Lake

Cloud Infrastructure: Google Cloud · Docker Containers

TECHNOLOGY STACK

BACKEND & STREAMING

FastAPI · Apache Kafka · Redis Cache · Python · SQLAlchemy · Pydantic

MACHINE LEARNING & AI

MediaPipe · Random Forest Regressor · Gradient Boosting · Ridge Regression · DBSCAN · Gemini API

FRONTEND DEVELOPMENT

React Native · Expo · React · Vite · Recharts · Google Maps API

DATA & STORAGE

PostgreSQL · PostGIS · Google Cloud Storage

SECURITY & DEVOPS

Firebase Authentication · Docker · HTTPS · TLS · RBAC

EXPECTED OUTCOMES

Fewer Idle Drivers

Proactive zone deployment eliminates wasted mileage before demand peaks

Lower Emissions

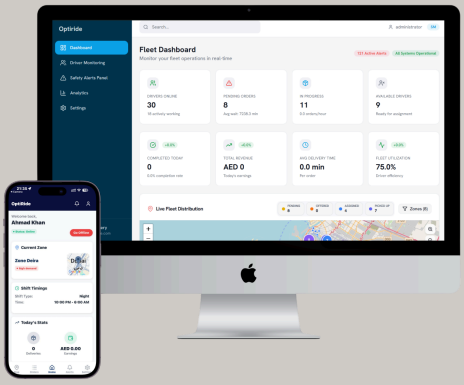
Optimised routing cuts unnecessary fuel use and CO_2 emission per delivery

Safer Roads

Fatigue and accident alerts reach drivers and admins before incidents escalate

Smarter Operations

AI insights give fleet managers actionable data instead of guesswork



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